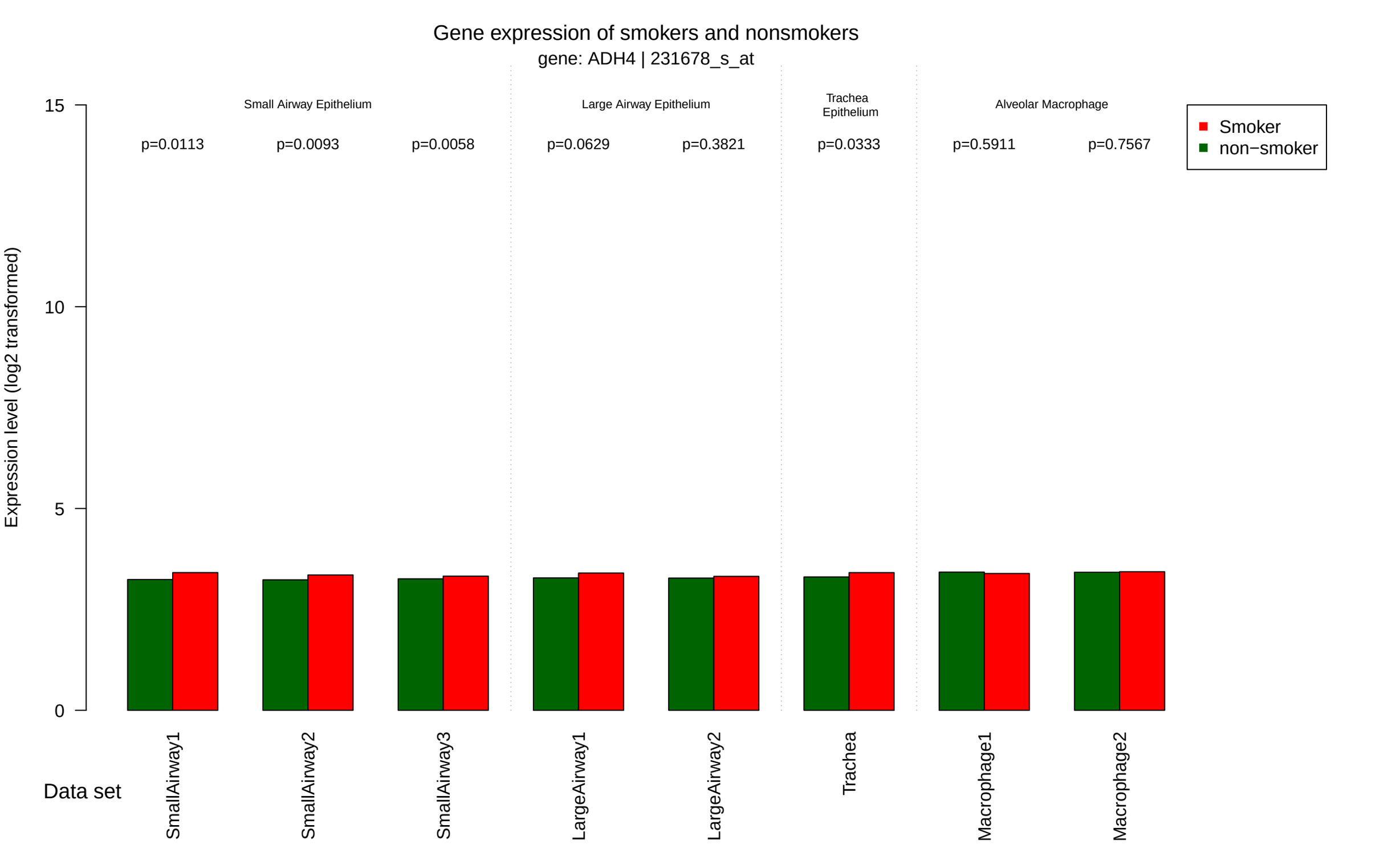
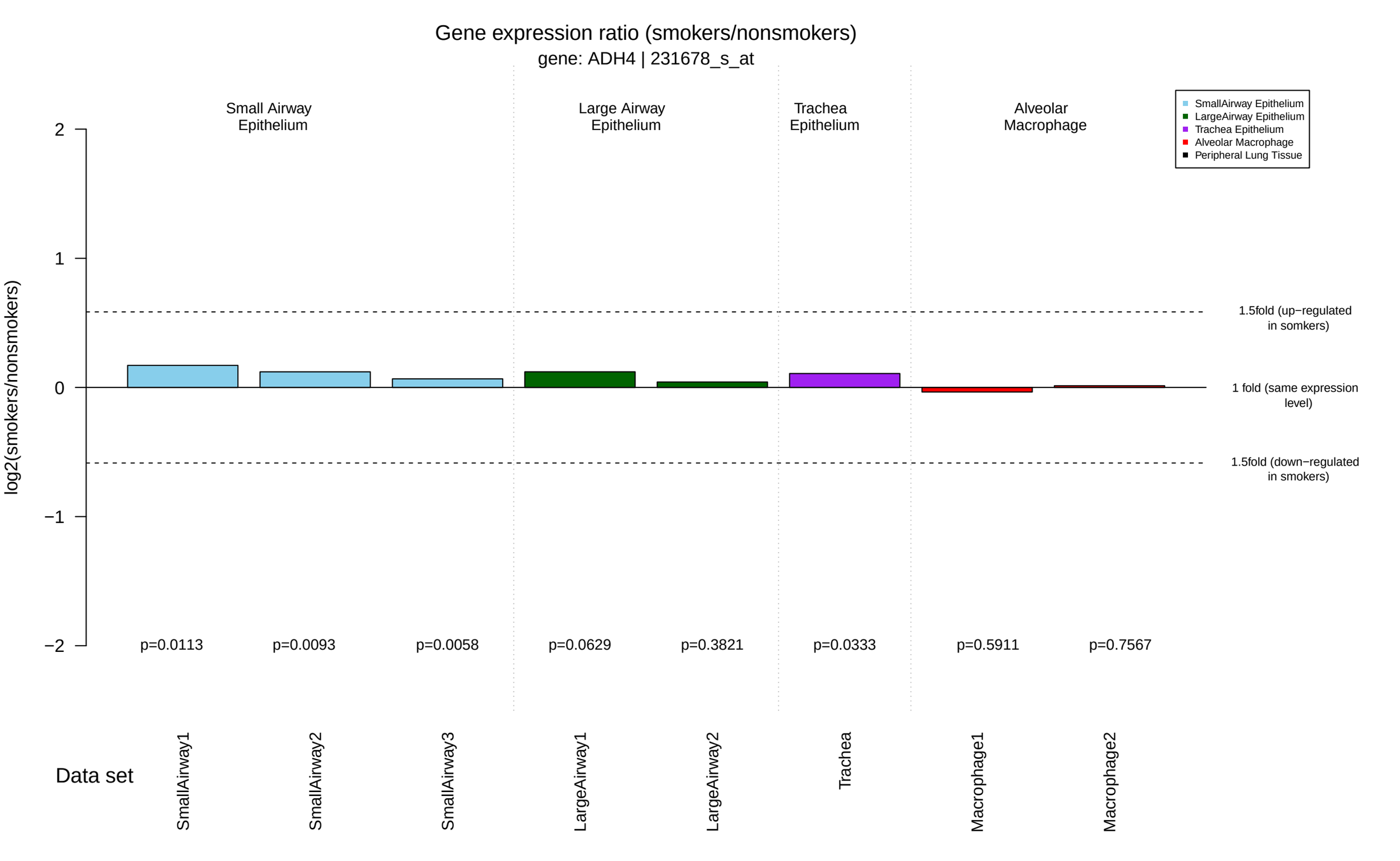
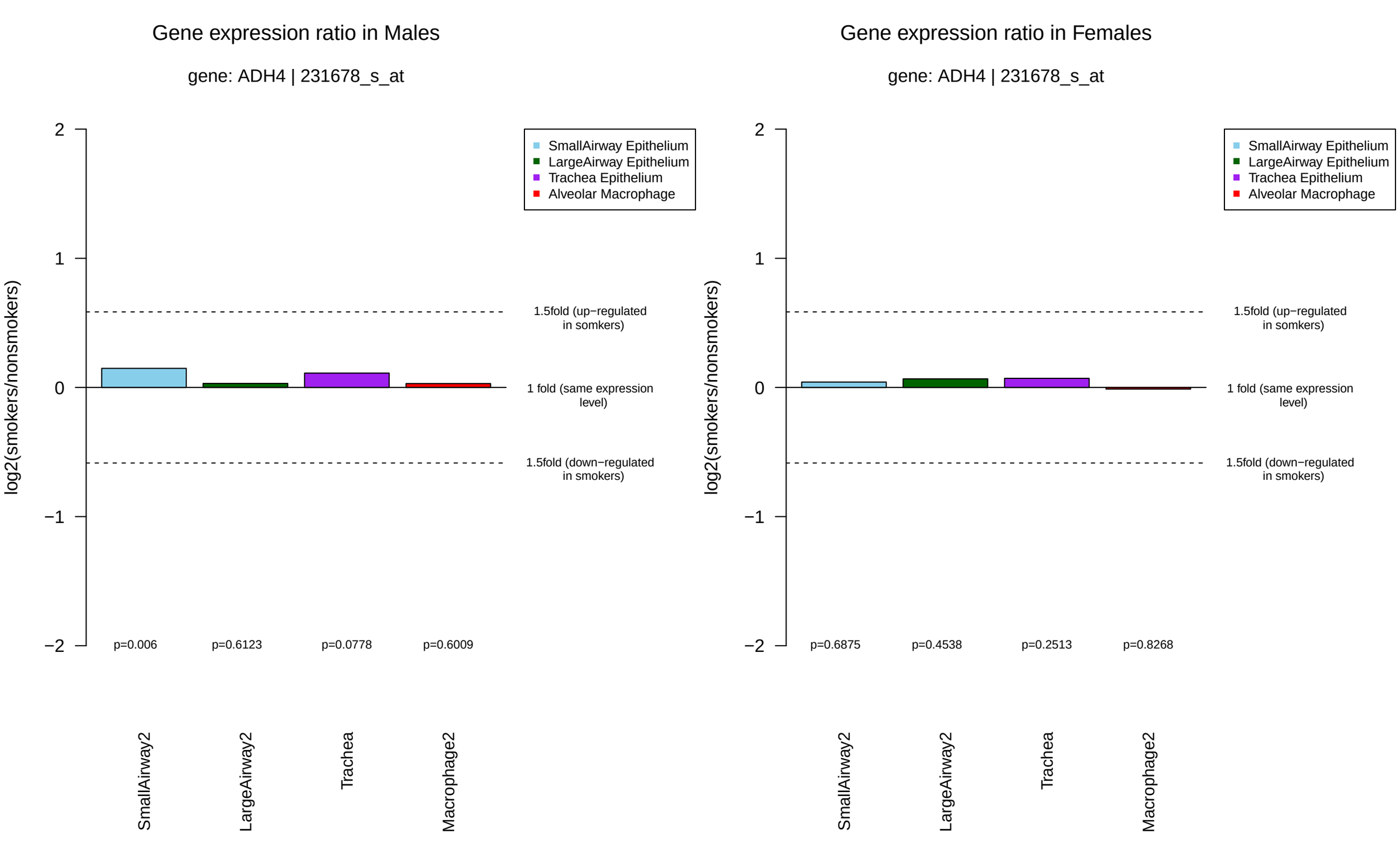
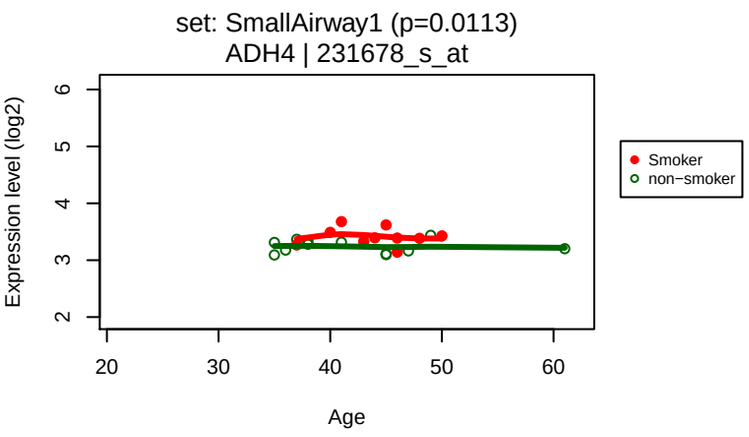
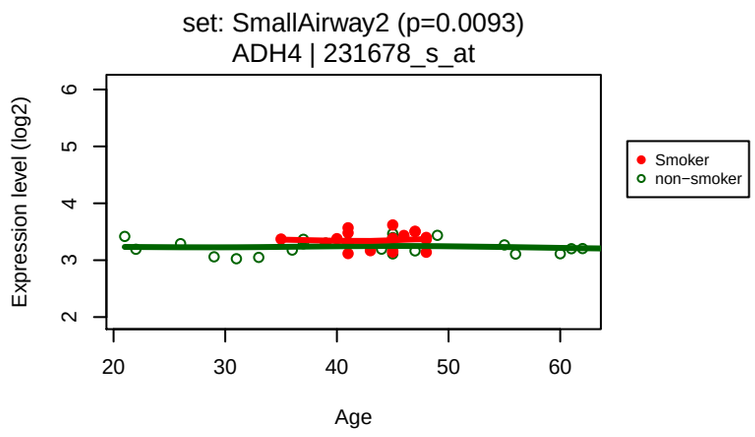
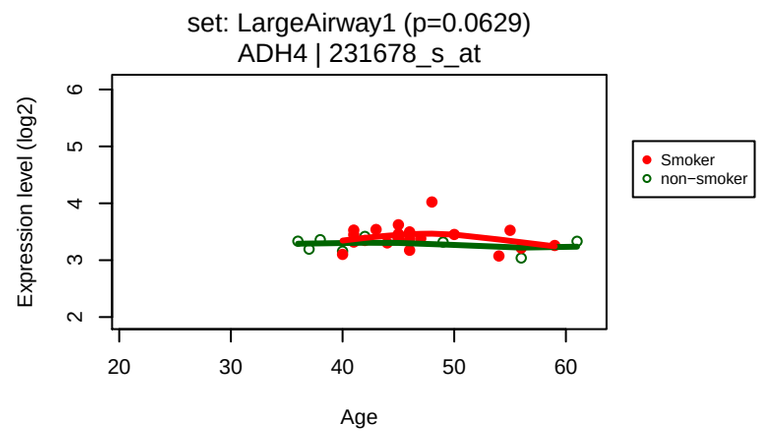

Gene expression of smokers and nonsmokers
gene: ADH4 | 231678_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0113
p=0.0093
p=0.0058
p=0.0629
p=0.3821
p=0.0333
p=0.5911
p=0.7567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

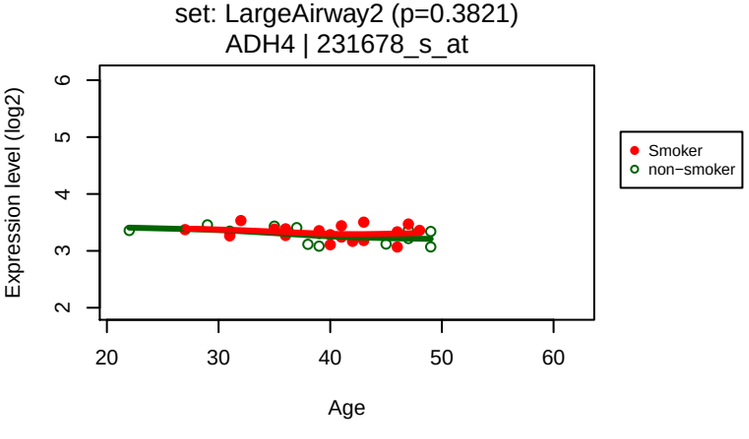

Gene expression ratio (smokers/nonsmokers)
gene: ADH4 | 231678_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0113
p=0.0093
p=0.0058
p=0.0629
p=0.3821
p=0.0333
p=0.5911
p=0.7567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

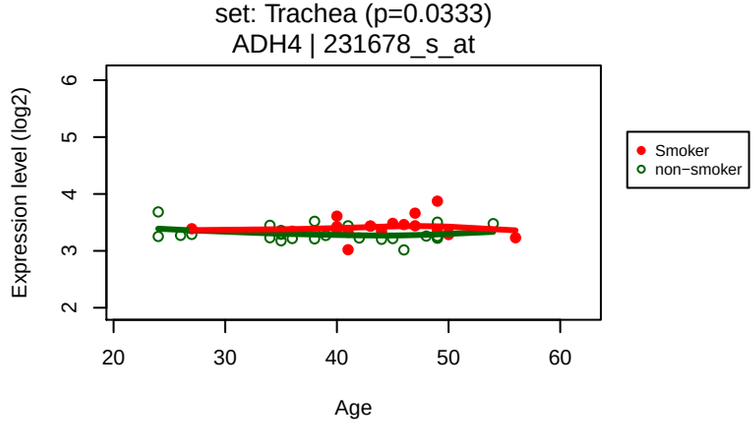

Gene expression ratio in Males
gene: ADH4 | 231678_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.006
p=0.6123
p=0.0778
p=0.6009
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADH4 | 231678_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6875
p=0.4538
p=0.2513
p=0.8268
SmallAirway2
LargeAirway2
Trachea
Macrophage2

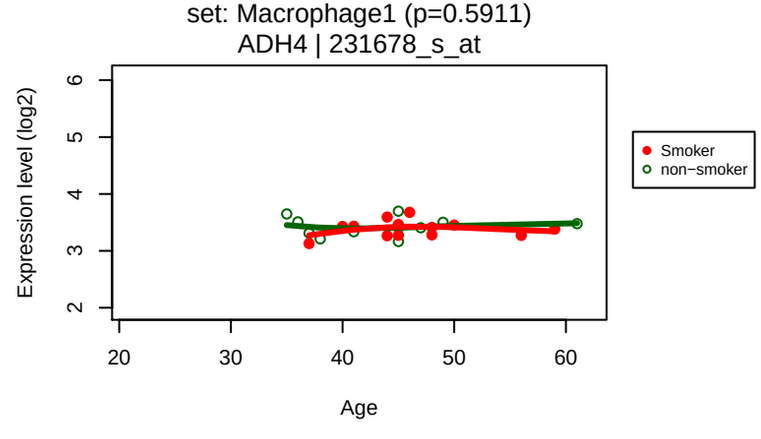

set: SmallAirway1 (p=0.0113)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker

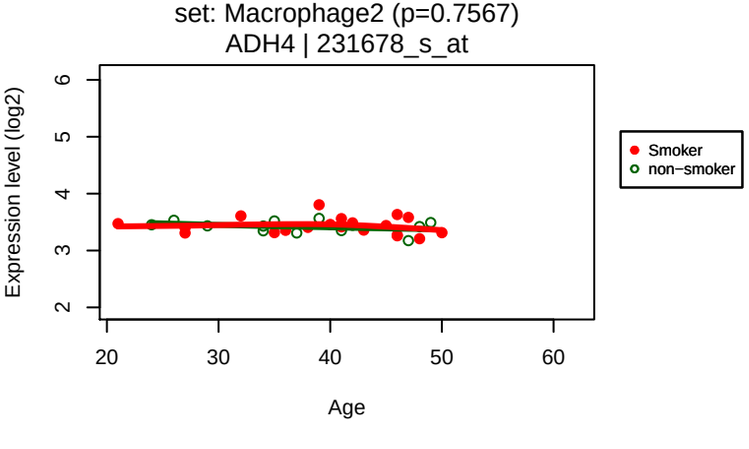

set: SmallAirway2 (p=0.0093)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker

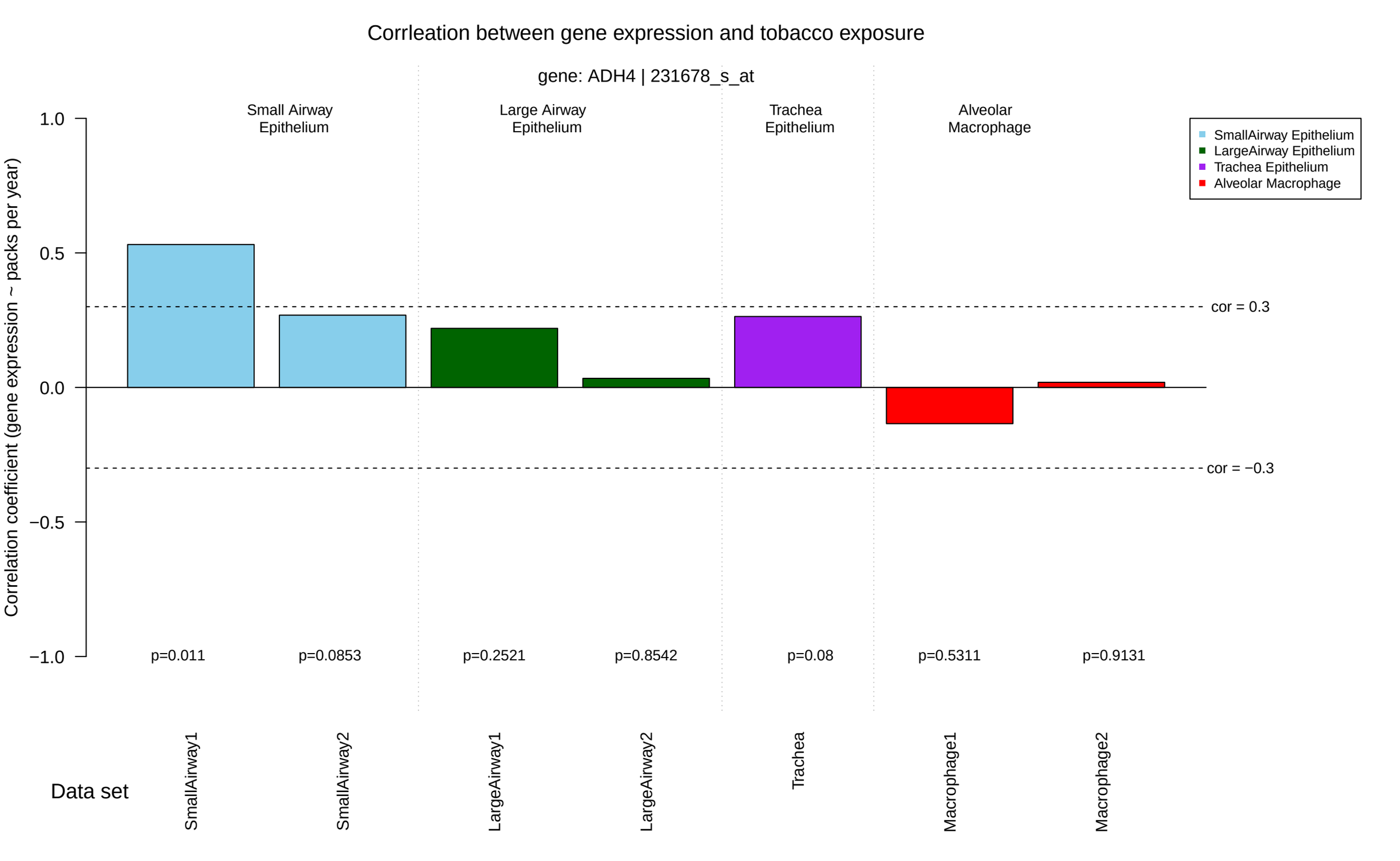

set: LargeAirway1 (p=0.0629)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3821)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0333)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5911)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7567)
ADH4 | 231678_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADH4 | 231678_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.011
p=0.0853
p=0.2521
p=0.8542
p=0.08
p=0.5311
p=0.9131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2